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Constructing chiral aza-quaternary carbon centers by enantioselective carbonylative Heck reaction of *o*-iodoanilines with allenes†

Zhengshuai Xu, Chaoren Shen, Hongru Zhang, Peng Wang and Kaiwu Dong *

The construction of chiral aza-quaternary carbon centers *via* C–N bond formation has been achieved by a palladium-catalyzed asymmetric carbonylative Heck reaction of *o*-iodoanilines with allenes, providing chiral dihydroquinolinone derivatives with moderate to high yield and enantiomeric ratio (er). The er could be improved to 99/1 through the recrystallization of the isolated products. Adding *o*-iodoaniline into the reaction by syringe pump was crucial to diminish the side reactions of reactive *o*-iodoanilines and improve the chemoselectivity. Utilizing the chirality induction of an assembled aza-quaternary stereogenic center, molecules bearing vicinal chiral quaternary carbon centers or multiple chiral centers can be rapidly prepared.

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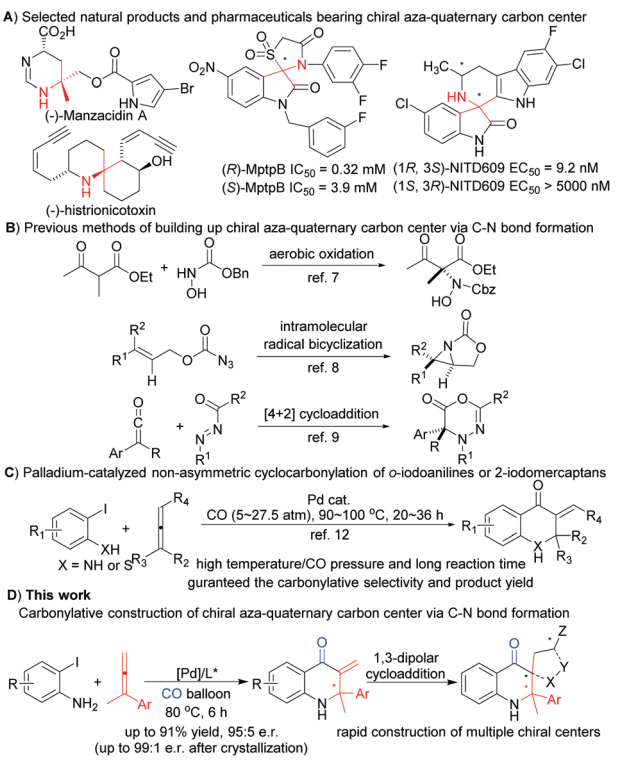
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The intrinsic properties of a chiral quaternary carbon center in terms of biological characteristics, *i.e.*, activity, metabolism, solubility, and hydrophobicity, endow the efficient catalytic methods for its construction with long-lasting research importance.[1] In particular, the crucial influence of the absolute configuration of the chiral aza-quaternary carbon center, which constitutes one of the ubiquitous structural units in natural products and pharmaceuticals (Scheme 1A),[2] on biological activity[3] gives significance to the building of such a stereocenter. In this context, diverse methods *via* C–C bond formation for preparing optically active molecules bearing an aza-quaternary carbon center have been well exploited. For example, the catalytic asymmetric addition of carbonic nucleophiles onto ketimines[2b,4] and the enantioselective substitution of carbonic electrophiles with α-nitrogen substituted pro-chiral carbonic nucleophiles[5] are commonly used protocols. In sharp contrast, catalytic asymmetric approaches *via* C–N bond formation have been far less explored, although they could provide direct and convenient access to these chiral amines. As we know, stereoselective protocols include aza-Michael addition,[6] aerobic oxidation of β-ketocarbonyls,[7] intramolecular radical bicyclization,[8] and cycloaddition[9] (Scheme 1B).

To meet the demand for accessing structurally complex chiral amines from comparatively common and inexpensive starting materials, diversifying the catalytic approaches to a



Scheme 1 Utility of aza-quaternary stereogenic center and related synthetic methods.

Chang-Kung Chuang Institute, Shanghai Key Laboratory of Green Chemistry and Chemical Process, School of Chemistry and Molecular Engineering, East China Normal University, Shanghai 200241, China. E-mail: kwdong@chem.ecnu.edu.cn

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chiral aza-quaternary carbon center *via* C–N bond formation is imperative.[10] Meanwhile, among the above-mentioned methods of access, there is still a lack of effective methods with the capability of synchronously forming multiple bonds. As a powerful strategy to achieve the simultaneous formation of C–N and C–C bonds across alkenes, asymmetric carboamination is a potent but almost unexplored C–N bond formation route to efficiently assemble a chiral aza-quaternary carbon center,[11] and its potential has yet to be exploited.[12]

Similar to the merits of carboamination, the palladium-catalysed carbonylative Heck reaction[13] also possesses the capability of simultaneously constructing two carbon–carbon bonds, and has been recognized as an important tool in the construction of α,β-unsaturated carbonyl compounds[14] and heterocycles.[15] Considering the common features of carboamination and the palladium-catalysed carbonylative Heck reaction and inspired by the work of Alper *et al.* on palladium-catalysed multicomponent achiral cyclocarbonylation under high CO pressure at 90–100 °C (Scheme 1C),[12] we intended to utilize a combination of these two powerful approaches to diversify the access to a chiral aza-quaternary stereogenic center *via* C–N bond formation and elevate the synthetic efficiency with a more active chiral palladium catalyst under milder conditions that are beneficial to inhibiting background reactions and acquiring high enantioselectivity while still retaining good carbonylative selectivity (*e.g.* lower CO pressure/reaction temperature and shorter reaction time). Herein, we report a palladium-catalysed enantioselective carbonylative Heck reaction between *o*-iodoanilines and allenes involving an alkene carboamination step (Scheme 1D). A variety of dihydroquinolinone derivatives containing a chiral aza-quaternary carbon center and one convertible alkene functional group was afforded with moderate to high yield and enantiomeric ratio (e.r.). Based on this reaction, a set of efficient methods for the rapid construction of multiple chiral centers with good yield and diastereoselectivity was developed.

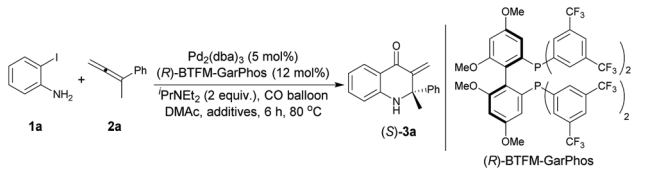
To explore the feasibility of synchronously assembling an aza-quaternary carbon stereocenter and accomplishing the formation of multiple C–C and C–N bonds, 2-iodoaniline **1a** and allene **2a** were selected as model substrates for a palladium-catalysed enantioselective carbonylative Heck reaction. Initially, a series of monodentate and bidentate privileged chiral ligands (29 chiral ligands) with different coordinating atoms was tested by employing tris(dibenzylideneacetone) dipalladium(0) as a palladium source and *N*,*N*-diisopropylethylamine as an acid scavenger in *N*,*N*-dimethylacetamide solution under CO atmosphere at 80 °C for 24 h (for detailed results see Table S1 in ESI†). Among these ligands, only axially bidentate phosphines, such as BINAP and substituent-modified MeO-BIPHEP ligands, could obtain the desired annulation product **3a** in either moderate yield or medium enantioselectivity. The demands for good product yield and high enantioselectivity failed to be met at the same time. Considering the relatively high yield obtained by (*R*)-BINAP (88%) and the relatively good enantioselectivity using (*R*)-BTFM-GarPhos (86 : 14 e.r.), we employed these two ligands to further explore the optimal conditions by solvent screening, tuning the Pd/P molar ratio, and CO pressure (Tables S2–S5 in ESI†). However, neither the yield nor the er was significantly improved. The palladium-to-phosphorus ratio has a sophisticated but noticeable impact on the yield of **3a**, *i.e.*, a lower Pd/P ratio (10 mol% Pd loading) diminished the yield of **3a** (for a detailed comparison see Table S5 in ESI†). During the course of optimization, the prevailing imbalances between the yield of **3a** and the conversion of substrate **1a** as well as the unequal conversion of substrates **1a** and **2a** were observed. Specifically, in almost all circumstances when adding one equivalent of **1a** and two equivalents of **2a**, **1a** was quantitatively consumed and more than one equivalent of **2a** was detected after the reaction, wherein product **3a** was obtained in only low to moderate yield. Compared with the reported Heck cyclization of EWG-substituted cumulated dienes with similar iodo substrates,[16] allene **1a** is much less activated due to the absence of EWG.[17] No GC-detectable or TLC-observable side products from the inter/intramolecular aminocarbonylation of **1a** (*e.g.* isatin[18] or 2-amino-*N*-(2-iodophenyl)-benzamide[19]) were identified. When 2-iodoaniline was replaced by the less reactive 2-bromoaniline or 2-amino phenyltriflate, no conversion of these substrates was observed (for detailed results see Table S7 in ESI†). Even in the absence of phosphine ligands, 2-iodoaniline could be quantitatively consumed under a CO atmosphere with $Pd_2(dba)_3$, PdI_2 or Pd/C, but only a little racemic product **3a** was generated (Table S6 in ESI†). These experimental results suggest that 2-iodoaniline is apparently much more reactive than allene **2a** in the presence of a palladium catalyst, which leads to the swift consumption of substrate **1a** through complicated side reactions.

Based on the above findings, we conjectured that keeping the concentration of substrate low in the reaction might prevent these side reactions resulting from the high reactivity of **1a** and thus attain better chemoselectivity. Moreover, transforming the neutral palladium intermediates to the cationic palladium species by silver salt additives clearly exhibited benefits to the chemoselectivity and enantioselectivity for the asymmetric Heck reaction.[20] Therefore, both altering the manner of feeding the substrate into the reaction solution and adding a silver salt were examined (Table 1). Compared to the result of adding the substrate in a one-portion manner (entry 1), adding **1a** into the reaction solution in a dropwise manner by a syringe pump over 6 hours indeed elevated the yield of product **3a** and preserved the good enantioselectivity (entry 2). Utilizing $AgBF_4$ as the additive and feeding **1a** into the reaction solution by programmed syringe pump over 6 hours could reduce the demand of allene **2a** and simultaneously obtained a much higher yield of **3a** than in the one-portion addition manner (entry 3 *versus* entry 4). Accelerating the rate of addition led to a decline in product yield (entry 5). Using AgOTf instead of $AgBF_4$ and adding **1a** by syringe pump could also have the effect of drastically inhibiting the undesirable side reaction (entry 6), but the yield and enantioselectivity are slightly inferior to those obtained using $AgBF_4$. These results clearly exhibited the usefulness of utilizing a syringe pump in

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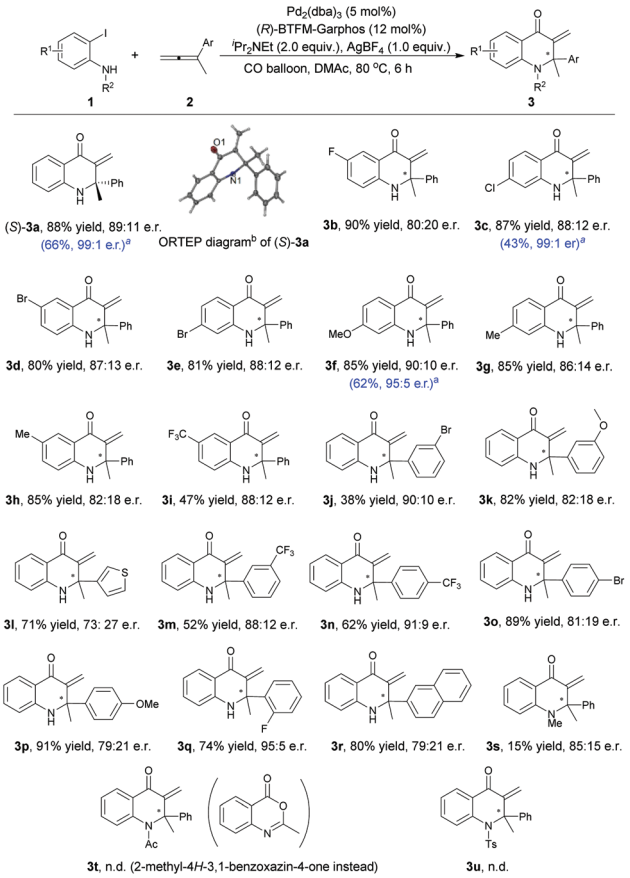
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Table 1 Optimization of reaction conditions



Entry	Additive	Yield (%)	e.r.
1	—	34	90 : 10
2[a]	—	50	89 : 11
3	$AgBF_4$	29	92 : 8
4[a]	$AgBF_4$	96 (88[b], 66[c])	89 : 11 (99 : 1[c])
5[d]	$AgBF_4$	90	88 : 12
6[a]	AgOTf	95	85 : 15

Unless otherwise noted, reaction conditions: **1a** (0.2 mmol, 1.0 equiv.), **2a** (0.4 mmol, 2.0 equiv.), $Pd_2(dba)_3$ (0.01 mmol, 5 mol%), (*R*)-BTFM-GarPhos (0.024 mmol, 12 mol%), $^iPr_2NEt$ (0.4 mmol, 2.0 equiv.), additive (0.2 mmol, 1.0 equiv.), DMAc (1.0 mL), 6 h. Yield was determined by GC using decane as internal standard and the enantiomeric ratio (er) was determined by chiral HPLC. [a] **1a** was dissolved in 0.5 mL DMAc and fed into the reaction solution (0.5 mL DMAc dissolving **2a** (0.3 mmol, 1.5 equiv.), $Pd_2(dba)_3$, (*R*)-BTFM-GarPhos, $^iPr_2NEt$ and additive) with a programmable syringe pump at 80 °C under atmospheric CO pressure over 6 h. [b] Isolated yield. [c] After recrystallization with petroleum ether/dichloromethane. [d] **1a** was dissolved in 0.5 mL DMAc and fed into the reaction solution (0.5 mL DMAc dissolving **2a** (0.3 mmol, 1.5 equiv.), $Pd_2(dba)_3$, (*R*)-BTFM-GarPhos $^iPrNEt_2$ and additive) with a programmable syringe pump at 80 °C under atmospheric CO pressure over 2 h, then further reacted for 4 h. DMAc = *N*,*N*-dimethylacetamide.



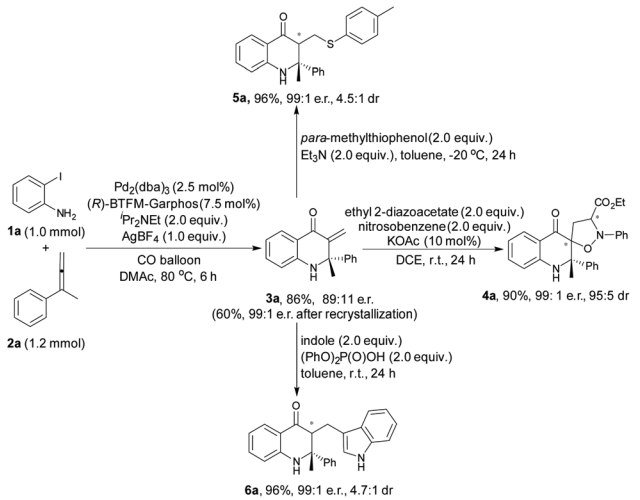
Scheme 2 Substrate scope of *o*-iodoanilines and allenes. Unless otherwise noted, reaction conditions: **1** (0.2 mmol, 1.0 equiv.), **2** (0.3 mmol, 1.5 equiv.), $Pd_2(dba)_3$ (0.01 mmol, 5 mol%), (*R*)-BTFM-Garphos (0.024 mmol, 12 mol%), $^iPr_2NEt$ (0.4 mmol, 2.0 equiv.), $AgBF_4$ (0.2 mmol, 1.0 equiv.), CO balloon, DMAc (1.0 mL), **1** was dissolved in 0.5 mL DMAc and fed into the reaction solution (0.5 mL DMAc) with a programmable syringe pump over 6 h. Isolated yields. The enantiomeric ratio (e.r.) was determined by chiral HPLC. [a] After recrystallization with petroleum ether/dichloromethane. [b] Thermal ellipsoids set at the 50% probability level. dba = dibenzylideneacetone; DMAc = *N*,*N*-dimethylacetamide.

eliminating the side reactions encountered in the batch-type carbonylative reaction. Gratifyingly, enantiomeric enrichment was observed for the carbonylation product, and **3a** was afforded as a solid with 99 : 1 e.r. and 66% yield after recrystallization with petroleum ether/dichloromethane.

With the optimized reaction conditions in hand, we investigated the substrate scope of *o*-iodoanilines and allenes (Scheme 2). The carbonylative Heck reaction between *o*-iodoanilines **1** and allenes **2** bearing different functional groups gave chiral methylene-2,3-dihydro-1*H*-quinolin-4-ones in moderate to excellent yields with a good enantiomeric ratio. Both electron-rich and electron-deficient substitutes on the phenyl ring of *o*-iodoaniline substrate **1** or 3-phenyl 1,2-butadiene **2** are well tolerated (**3b–3k**, **3m–3q**). The bromo and chloro groups on the phenyl ring of the substrates remained intact by this protocol and the corresponding carbonylative Heck reaction products were afforded in moderate to good yields with a good enantiomeric ratio (**3c–3e**, **3j**, **3o**). These halogen substituents provide potential reaction sites for the further transformation of products *via* transition metal-catalysed cross coupling. The thienyl substituent in the allene substrate (**3l**) was also compatible with this protocol. Switching the phenyl group on the allene substrate to a fused aryl ring (**3r**) had little impact on yield or enantioselectivity. Attaching a methyl substituent onto the amino group of *o*-iodoaniline largely reduced its reactivity, which led to the poor conversion and low yield of **3s**. Modifying the amino group of *o*-iodoaniline with acetyl resulted in intramolecular cyclocarbonylation forming 2-methyl-4*H*-3,1-benzoxazin-4-one,[21] while a series of inseparable side products was generated for *N*-tosyl *o*-iodoaniline. Again, we found that the major enantiomer of carbonylative annulation products could be further enriched to higher optical purity by recrystallizing the isolated reaction products (*e.g.* **3c** and **3f**). The absolute configuration of **3a** was determined to be (*S*) by X-ray crystallographic analysis.

Considering the distinct effect of the substrate feeding manner on the product yield and to further prove the reliability of this protocol, the scale-up of this carbonylative Heck reaction between substrates **1a** and **2a** was examined and it showed that the good chemoselectivity and enantioselectivity were unaffected at a larger reaction scale (1.0 mmol) with less catalyst loading (Scheme 3). By recrystallizing the isolated product in a mixed solvent of petroleum ether and dichloromethane, the chiral carbonylative product **3a** with high optical

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Scheme 3 Scale-up reaction and construction of vicinal or multiple chiral centers *via* chirality induction.

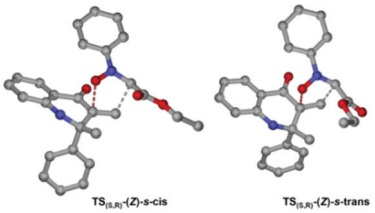


Fig. 1 3D structures of the two most kinetically preferred transition states for the 1,3-dipolar cycloaddition of the *in situ* formed nitrone intermediate onto (*S*)-configuration alkene **3a** revealed by DFT computation (all hydrogen atoms are omitted for clarity).

purity (99 : 1 e.r.) was obtained in 60% yield (Scheme 3). More than building up the aza-quaternary stereogenic center, the other noteworthy feature of this palladium-catalysed asymmetric carbonylative Heck reaction is the generation of a disubstituted terminal alkene functional group, which is not only conjugated to the implanted carbonyl group, but is also adjacent to the aza-quaternary carbon stereocenter. It provides a valuable reaction site to produce vicinal or multiple chiral centers by asymmetric induction relying on the intervention of the aza-quaternary carbon stereocenter. To validate this design of readily constructing vicinal or multiple chiral centers within two steps, we selected the asymmetric three-component 1,3-dipolar cycloaddition of readily accessible ethyl 2-diazoacetate, nitrosobenzene with alkene **3a**,[22] as well as the asymmetric Michael addition of *para*-methylthiol or indole nucleophile on Michael acceptor **3a**[23] (Scheme 3). To our delight, through the cycloaddition of an *in situ* formed active nitrone intermediate onto the alkene group of **3a**, spiromethylene-2,3-dihydro-1*H*-quinolin-4-one isoxazolidine **4a**, which possesses three chiral stereogenic centers including a pair of vicinal aza-quaternary and oxa-quaternary stereocenters, could be obtained in high yield with good enantioselectivity and diastereoselectivity under mild conditions. By Michael addition onto the carbonyl conjugated C–C double bond of **3a**, chiral thioether **5a** and chiral indole derivative **6a**, both bearing a pair of vicinal chiral centers, were afforded in good yield with excellent enantiomeric ratio and good diastereoselectivity (Scheme 3).

Owing to the difficulty that 1,3-dipolar cycloaddition product **4a** was rather liable to form a non-crystalline solid even by various crystallization techniques, the effort to determine the configuration of three chiral centers in **4a** by X-ray diffraction failed. Thus, we employed density functional theory (DFT) calculations to predict the configuration and elucidate the high diastereoselectivity of this 1,3-dipolar cycloaddition. Since the *in situ* formed active nitrone intermediate 1-ethylcarboxy *N*-phenylmethanimine oxide has four different isomers consisting of (*E*)- or (*Z*)-configurations on the C–N double bond and s-*cis* or s-*trans* conformations[24] and pondering these two factors: (i) the equilibrium between these isomers is dependent on the polarity of solvent[25] and the presence of Lewis acids;[26] (ii) product **4a** has eight potential enantiomers and diastereomers, we searched out a total of sixteen 1,3-dipolar cycloaddition transition states for the addition of nitrone intermediate 1-ethylcarboxy *N*-phenylmethanimine oxide onto the alkene group neighbouring the (*S*)-configuration quaternary center in **3a** (for more details see ESI†). By comparing the potential energy location of these transition states in the solvation of dichloroethane, we found that two transition states, both of which are towards generating an (*S*)-configuration oxa-quaternary carbon center and (*R*)-configuration aza-tertiary carbon center (Fig. 1), are much more energetically advantageous than the other fourteen transition states (for detailed results and analysis see ESI†). This outcome is highly consistent with the observed high enantioselectivity and diastereoselectivity of cycloaddition product **4a** and revealed the good chiral induction contributed from the carbonylation-implanted aza-quaternary stereocenter. The successful implementation of these conversions proves the potent utilization of an aza-quaternary stereocenter assembled through this palladium-catalysed asymmetric carbonylative Heck reaction in the rapid construction of multiple chiral centers.

In summary, an efficient method for the construction of a chiral aza-quaternary carbon center *via* C–N bond formation by a palladium-catalysed asymmetric carbonylative Heck reaction of *o*-iodoanilines with allenes was developed. Adopting the dropwise addition manner in this batch-type reaction could effectively inhibit the proceeding of side reactions due to the high reactivity of the *o*-iodoaniline substrate. By utilizing the aza-quaternary stereogenic center to impose chirality induction in the subsequent alkene transformation, highly functionalized and otherwise inaccessible molecules containing a pair of vicinal chiral centers can be facilely prepared.

Conflicts of interest

There are no conflicts to declare.

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Notes and references

1 (*a*) E. A. Peterson and L. E. Overman, Contiguous stereogenic quaternary carbons: A daunting challenge in natural products synthesis, *Proc. Natl. Acad. Sci. U. S. A.*, 2004, **101**, 11943–11948; (*b*) P. G. Cozzi, R. Hilgraf and N. Zimmermann, Enantioselective Catalytic Formation of Quaternary Stereogenic Centers, *Eur. J. Org. Chem.*, 2007, 5969–5994; (*c*) K. W. Quasdorf and L. E. Overman, Catalytic enantioselective synthesis of quaternary carbon stereocenters, *Nature*, 2014, **516**, 181–191; (*d*) J. Feng, M. Holmes and M. J. Krische, Acyclic Quaternary Carbon Stereocenters via Enantioselective Tran-sition Metal Catalysis, *Chem. Rev.*, 2017, **117**, 12564–12580; (*e*) F. Zhou, L. Zhu, B.-W. Pan, Y. Shi, Y.-L. Liu and J. Zhou, Catalytic enantioselective construction of vicinal quaternary carbon stereocenters, *Chem. Sci.*, 2020, **11**, 9341–9365.

2 (*a*) A. Hager, N. Vrielink, D. Hager, J. Lefranc and D. Trauner, Synthetic approaches towards alkaloids bearing α-tertiary amines, *Nat. Prod. Rep.*, 2016, **33**, 491–522; (*b*) K. Bera and I. N. N. Namboothiri, Asymmetric Synthesis of Quaternary α-Amino Acids and Their Phosphonate Analogues, *Asian J. Org. Chem.*, 2014, **3**, 1234–1260; (*c*) H. Jiang, Y. Jin and J. Lin, New Progress in Asymmetric Synthesis of Quaternary α-Amino Acids, *Mini-Rev. Org. Chem.*, 2017, **14**, 434–447.

3 (*a*) A. K. Chatterjee and B. K. S. Yeung, Back to the Future: Les-sons Learned in Modern Target-based and Whole-Cell Lead Optimization of Antimalarials, *Curr. Top. Med. Chem.*, 2012, **12**, 473–483; (*b*) V. V. Vintonyak, K. Warburg, H. Kruse, S. Grimme, K. Hübel, D. Rauh and H. Waldmann, Identification of Thiazolidinones Spiro-Fused to Indolin-2-ones as Potent and Selective Inhibitors of the Mycobacterium tuberculosis Protein Tyrosine Phosphatase B, *Angew. Chem., Int. Ed.*, 2010, **49**, 5902–5905.

4 (*a*) O. Riant and J. Hannedouche, Asymmetric catalysis for the construction of quaternary carbon centers: nucleophilic addition on ketones and ketimines, *Org. Biomol. Chem.*, 2007, **5**, 873–888; (*b*) B. EftekhariSis and M. Zirak, α-Imino Esters in Organic Synthesis: Recent Advances, *Chem. Rev.*, 2017, **117**, 8326–8419; (*c*) R. Wada, T. Shibuguchi, S. Makino, K. Oisaki, M. Kanai and M. Shibasaki, Catalytic Enantioselective Allylation of Ketoimines, *J. Am. Chem. Soc.*, 2006, **128**, 7687–7691; (*d*) R. Shintani, M. Takeda, T. Tsuji and T. Hayashi, Rhodium-Catalyzed Asymmetric Arylation of N-Tosyl Ketimines, *J. Am. Chem. Soc.*, 2010, **132**, 13168–13169; (*e*) Y.-H. Wang, J.-S. Tian, P.-W. Tan, Q. Cao, X.-X. Zhang, Z.-Y. Cao, F. Zhou, X. Wang and J. Zhou, Regiodivergent Intramolecular Nucleophilic Addition of Ketimines for the Diverse Synthesis of Azacycles, *Angew. Chem., Int. Ed.*, 2020, **59**, 1634–1643.

5 R. Kuwano and Y. Ito, Catalytic Asymmetric Allylation of Prochiral Nucleophiles, α-Acetamido-β-ketoesters, *J. Am. Chem. Soc.*, 1999, **121**, 3236–3237.

6 (*a*) L. Wang, J. Chen and Y. Huang, Highly Enantioselective Aza-Michael Reaction between Alkyl Amines and β-Trifluoromethyl β-Aryl Nitroolefins, *Angew. Chem., Int. Ed.*, 2015, **54**, 15414–15418; (*b*) Z. Yang, Z. Wang, S. Bai, X. Liu, L. Lin and X. Feng, *Org. Lett.*, 2011, **13**, 596–599; (*c*) S. Ma, N. Jiao, Z. Zheng, Z. Ma, Z. Lu, L. Ye, Y. Deng and G. Chen, Cu- and Pd-Catalyzed Asymmetric One-Pot Tandem Addition-Cyclization Reaction of 2-(2′,3′-Alkadienyl)-β-keto Esters, Organic Halides, and Dibenzyl Azodicarboxylate: An Effective Protocol for the Enantioselective Synthesis of Pyrazolidine Derivatives, *Org. Lett.*, 2004, **6**, 2193–2196.

7 C. Xu, L. Zhang and S. Luo, Merging Aerobic Oxidation and Enamine Catalysis in the Asymmetric α-Amination of β-Ketocarbonyls Using N-Hydroxycarbamates as Nitrogen Sources, *Angew. Chem., Int. Ed.*, 2014, **53**, 4149–4153.

8 H. Jiang, K. Lang, H. Lu, L. Wojtas and X. P. Zhang, Asymmetric Radical Bicyclization of Allyl Azidoformates via Cobalt(II)-Based Metalloradical Catalysis, *J. Am. Chem. Soc.*, 2017, **139**, 9164–9167.

9 (*a*) D. M. Dalton and T. Rovis, Catalytic, Asymmetric Indolizidi-none Aza-Quaternary Stereocenter Synthesis: Expedient Synthesis of the Cylindricine Alkaloid Core, *Org. Lett.*, 2013, **15**, 2346–2349; (*b*) X.-L. Huang, L. He, P.-L. Shao and S. Ye, [4 + 2] Cycloaddition of Ketenes with N-Benzoyldiazenes Catalyzed by N-Heterocyclic Carbenes, *Angew. Chem., Int. Ed.*, 2009, **48**, 192–195.

10 A. K. Mailyan, J. A. Eickhoff, A. S. Minakova, Z. Gu, P. Lu and A. Zakarian, Cutting-Edge and Time-Honored Strategies for Ste-reoselective Construction of C-N Bonds in Total Synthesis, *Chem. Rev.*, 2016, **116**, 4441–4557.

11 (*a*) H. Jiang and A. Studer, Intermolecular radical carboamination of alkenes, *Chem. Soc. Rev.*, 2020, **49**, 1790–1811; (*b*) J. He, Y. Xue, B. Han, C. Zhang, Y. Wang and S. Zhu, Nickel-Catalyzed Asymmetric Reductive 1,2-Carboamination of Unactivated Alkenes, *Angew. Chem., Int. Ed.*, 2020, **59**, 2328–2332; (*c*) D. R. White, J. T. Hutt and J. P. Wolfe, Asymmetric Pd-Catalyzed Alkene Carboamination Reactions for the Synthesis of 2-Aminoindane Derivatives, *J. Am. Chem. Soc.*, 2015, **137**, 11246–11249.

12 (*a*) F. Ye and H. Alper, Ionic-Liquid-Promoted Palladium-Catalyzed Multicomponent Cyclocarbonylation of o-Iodoanilines and Allenes To Form Methylene-2,3-dihydro-1H-quinolin-4-ones, *J. Org. Chem.*, 2007, **72**, 3218–3222; (*b*) W.-J. Xiao and H. Alper, Regioselective Carbonylative Heteroannulation of o-Iodothiophenols with Allenes and Carbon Monoxide Catalyzed by a Palladium Complex: A Novel and Efficient Access to Thiochroman-4-one Derivatives, *J. Org. Chem.*, 1999, **64**, 9646–9652.

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13 Carbonylative Heck reaction (*i.e.*, the Heck reactions between the Pd-acyl intermediate and alkene) should not be confused with the so-called Heck carbonylation, a term which is sometimes used for the palladium-catalysed Heck reaction terminated by the carbonylation of Pd-alkyl intermediates. Relevant examples see: (*a*) J.-B. Peng, Recent Advances in Carbonylative Difunctionalization of Alkenes, *Adv. Synth. Catal.*, 2020, **362**, 3059–3080; (*b*) T. Hayashi, J. Tang and K. Kato, Palladium-Catalyzed Enantioselective Carbonylative Cyclization of Aryl and Alkenyl Triflates with Carbon Monox-ide, *Org. Lett.*, 1999, **1**, 1487–1489; (*c*) R. C. Carmona, O. D. Koster and C. R. D. Correia, Chiral N,N Ligands Enabling Palladium-Catalyzed Enantioselective Intramolecular Heck-Matsuda Carbonylation Reactions by Sequential Migratory and CO Insertions, *Angew. Chem., Int. Ed.*, 2018, **57**, 12067–12070; (*d*) H. Hu, F. Teng, J. Liu, W. Hu, S. Luo and Q. Zhu, Enantioselective Synthesis of 2-Oxindole Spirofused Lactones and Lactams by Heck/Carbonylative Cylization Sequences: Method Development and Applications, *Angew. Chem., Int. Ed.*, 2019, **58**, 9225–9229; (*e*) C. Cheng, B. Wan, B. Zhou, Y. Gu and Y. Zhang, Enantioselective synthesis of quaternary 3,4-dihydroisoquinolinones via Heck carbonylation reactions: development and application to the synthesis of Minalrestat analogues, *Chem. Sci.*, 2019, **10**, 9853–9858; (*f*) Z. Yuan, Y. Zeng, Z. Feng, Z. Guan, A. Lin and H. Yao, Constructing chiral bicy-clo[3.2.1]octanes via palladium-catalyzed asymmetric tandem Heck/carbonylation desymmetrization of cyclopentenes, *Nat. Commun.*, 2020, **11**, 2544–2551; (*g*) M. Chen, X. Wang, P. Yang, X. Kou, Z.-H. Ren and Z.-H. Guan, Palladium-Catalyzed Enantioselective Heck Carbonylation with a Monodentate Phosphoramidite Ligand: Asymmetric Synthesis of (+)-Physostigmine, (+)-Physovenine, and (+)-Folicanthine, *Angew. Chem., Int. Ed.*, 2020, **59**, 12199–12205.

14 (*a*) X.-F. Wu, H. Neumann, A. Spannenberg, T. Schulz, H. Jiao and M. Beller, Development of a General Palladium-Catalyzed Carbonylative Heck Reaction of Aryl Halides, *J. Am. Chem. Soc.*, 2010, **132**, 14596–14602; (*b*) X.-F. Wu, H. Neumann and M. Beller, Palladium-Catalyzed Coupling Reactions: Carbonylative Heck Reactions To Give Chalcones, *Angew. Chem., Int. Ed.*, 2010, **49**, 5284–5288; (*c*) P. Hermange, T. M. Gøgsig, A. T. Lindhardt, R. H. Taaning and T. Skrydstrup, Carbonylative Heck Reactions Using CO Generated ex Situ in a Two-Chamber System, *Org. Lett.*, 2011, **13**, 2444–2447; (*d*) J. Schranck, X.-F. Wu, H. Neumann and M. Beller, Palladium-Catalyzed Carbonylative Heck Reaction of Aryl Bromides with Vinyl Ethers to 3-Alkoxy Alkenones and Pyrazoles, *Chem. – Eur. J.*, 2012, **18**, 4827–4831.

15 (*a*) C. Shen and X.-F. Wu, Palladium-Catalyzed Carbonylative Multicomponent Reactions, *Chem. – Eur. J.*, 2017, **23**, 2973–2987; (*b*) X.-F. Wu, H. Neumann and M. Beller, Synthesis of Heterocycles via Palladium-Catalyzed Carbonylations, *Chem. Rev.*, 2013, **113**, 1–35; (*c*) X. Zhou, A. Chen, W. Du, Y. Wang, Y. Peng and H. Huang, Palladium-Catalyzed Hydrocarbonylative Cyclization Enabled by Formal Insertion of Aromatic C=N Bonds into Pd-Acyl Bonds, *Org. Lett.*, 2019, **21**, 9114–9118; (*d*) S. Zou, B. Gao, Y. Huang, T. Zhang and H. Huang, Palladium-Catalyzed Hydrocarbonylative Cyclization of 1,5-Dienes, *Org. Lett.*, 2019, **21**, 6333–6336; (*e*) X. Zhou, Y. Ding and H. Huang, Palladium-Catalyzed Carbonylative Difunctionalization of C=N Bond of Azaarenes or Imines to Quinazolinones, *Chem. – Asian J.*, 2020, **15**, 1678–1682; (*f*) K. Xiang, P. Tong, B. Yan, L. Long, C. Zhao, Y. Zhang and Y. Li, Synthesis of Benzannulated [6,6]-Spiroketals by a One-Pot Carbonylative Sonogashira Coupling/Double Annulation Reaction, *Org. Lett.*, 2019, **21**, 412–416.

16 (*a*) M. P. Pavan, M. Chakravarty and K. C. K. Swamy, Pd-Catalyzed Reactions of Allenylphosphonates and Related Allenes with Functionalized 2-Iodophenols, 2-Iodobenzoic Acid, and 2-Iodobenzyl Alcohol Leading to Functionalized Benzofurans, Isocoumarins, and Benzopyrans, *Eur. J. Org. Chem.*, 2009, 5927–5940; (*b*) M. N. Reddy and K. C. K. Swamy, pPapaerlladium-Catalyzed Reactions of Allenes with 2-Iodobenzenesulfonamides: Simple Synthesis of Benzosultams under Green Conditions, *Synthesis*, 2014, **46**, 1091–1099.

17 S. Ma, Some Typical Advances in the Synthetic Applications of Allenes, *Chem. Rev.*, 2005, **105**, 2829–2871.

18 S. R. Laursen, M. T. Jensen, A. T. Lindhardt, M. F. Jacobsen and T. A. Skrydstrup, Palladium-Catalyzed Double Carbonylation Approach to Isatins from 2-Iodoanilines, *Eur. J. Org. Chem.*, 2016, 1881–1885.

19 (*a*) L. Akerbladh and L. R. Odell, Synthesis of 2-Aminoquinazolinones via Carbonylative Coupling of ortho-Iodoanilines and Cyanamide, *J. Org. Chem.*, 2016, **81**, 2966–2973; (*b*) C. Shen, N. Y. T. Man, S. Stewart and X.-F. Wu, Palladium-catalyzed dicarbonylative synthesis of tetracycle quinazolinones, *Org. Biomol. Chem.*, 2015, **13**, 4422–4425.

20 A. Ashimori, B. Bachand, L. E. Overman and D. J. Poon, Catalytic Asymmetric Synthesis of Quaternary Carbon Centers. Explora-tory Investigations of Intramolecular Heck Reactions of (E)-α,β-Unsaturated 2-Haloanilides and Analogues To Form Enantioenriched Spirocyclic Products, *J. Am. Chem. Soc.*, 1998, **120**, 6477–6487.

21 (*a*) X.-F. Wu, J. Schranck, H. Neumann and M. Beller, A Convenient and General Palladium-Catalyzed Carbonylative Coupling for the Synthesis of 2-Arylbenzoxazinones, *Chem. – Eur. J.*, 2011, **17**, 12246–12249; (*b*) J. Salvadori, E. Balducci, S. Zaza, E. Petricci and M. Taddei, Microwave-Assisted Carbonylation and Cyclocarbonylation of Aryl Iodides under Ligand Free Heterogeneous Catalysis, *J. Org. Chem.*, 2010, **75**, 1841–1847; (*c*) C. Larksarp and H. Alper, A Simple Synthesis of 2-Substituted-4H-3,1-benzoxazin-4-ones by Palladium-Catalyzed Cyclocarbonylation of o-Iodoanilines with Acid Chlorides, *Org. Lett.*, 1999, **1**, 1619–1622.

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22 X. Li, T. Feng, D. Li, H. Chang, W. Gao and W. Wei, KOAc-Catalyzed One-Pot Three-Component 1,3-DipolarCycloaddition of α-Diazo Compounds, Nitrosoarenes, and Alkenes: An Approach to Functionalized Isoxazolidines, *J. Org. Chem.*, 2019, **84**, 4402–4412.

23 (*a*) Y.-P. Li, S.-F. Zhu and Q.-L. Zhou, Chiral Spiro Phosphoramide-Catalyzed Sulfa-Michael Addition/Enantioselective Protona-tion of Exocyclic Enones, *Org. Lett.*, 2019, **21**, 9391–9395; (*b*) Y.-P. Li, Z.-Q. Li, B. Zhou, M.-L. Li, X.-S. Xue, S.-F. Zhu and Q.-L. Zhou, Chiral Spiro Phosphoric Acid-Catalyzed Friedel-Crafts Conjugate Addition/Enantioselective Protonation Reactions, *ACS Catal.*, 2019, **9**, 6522–6529.

24 Y. Inouye, Y. Watanabe, S. Takahashi and H. Kakisawa, The Preparation of N-Benzyl-α-ethoxycarbonylnitrone and Its Reactions with Some Olefins, *Bull. Chem. Soc. Jpn.*, 1979, **52**, 3763–3764.

25 (*a*) Y. Inouye, J. Hara and H. Kakisawa, Novel E-Z Equilibrium of N-Alkyl-α-Alkoxycarbonylnitrone in Solution, *Chem. Lett.*, 1980, 1407–1410; (*b*) P. DeShong, C. M. Dicken, R. R. Staib, A. J. Freyer and S. M. Weinreb, Determination of Configuration and Conformation of Isoxazolidines by Nuclear Overhauser Effect Difference Spectroscopy, *J. Org. Chem.*, 1982, **47**, 4397–4403.

26 S. Kanemasa and T. Tsuruoka, Magnesium Bromide-Promoted *E*/*Z*-Isomerization of Carbonyl-Conjugated Nitrones and Highly Stereo- and Regioselective Cycloadditions to Allylic Alcohol Dipolarophiles, *Chem. Lett.*, 1995, 49–50.